Multiple Choice (10 marks)

1. How many distinct diagonals of a convex hexagon can be drawn?
   (a) 6
   (b) 8
   (c) 9
   (d) 36
2. Find the value of the expression 14 + (-71).
   (a) -85
   (b) 57
   (c) 85
   (d) -57
3. The original price of a new bicycle is \$138.00. If the bicycle is marked down 15%, what is the new price?
   (a) \$20.70
   (b) \$117.30
   (c) \$123.00
   (d) \$153.00
4. Suppose a, b, and c are positive numbers satisfying $a^2/b = 1, b^2/c = 2, c^2/a = 3$. Find a.
   (a) $12^{(}1/7)$
   (b) $7^{(}1/12)$
   (c) 1
   (d) 6
5. Roya paid \$48 for 12 cartons of orange juice. What is the unit rate per carton of orange juice that Roya paid?
   (a) \$3
   (b) \$4
   (c) \$6
   (d) \$12

True / False (10 marks)

Answer True or False

6. True or False: The number 336 is a powerful number, as it has prime factors that do not need to meet the square factor condition.

7. True or False: The maximum value of the function f(x) = xe$^{\{} - x\}ise.$

7. True or False: The quotient of the expression 2,314 divided by 4 is 578 with a remainder of 2.

8. True or False: The solution to the equation -47 = g + 24 is g = -23.

9. True or False: The value of the expression 28 x 42 is 1,176.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of finding all arithmetic sequences of consecutive odd integers that sum to 240. How many such sequences exist, and what methods did you use to determine this?

12. Calculate how many unique committees of 5 can be formed from a group of 15 people. Explain your reasoning and the mathematical principles involved in your solution.

End of Paper